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# Seeing every tree in the forest

Structure-aware and promptable tree segmentation  
in 3D forest LiDAR point clouds

Trung Thanh Nguyen

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## ABOUT ME

# Trung Thanh Nguyen



*M.Sc. graduation  
Nagoya University, Japan (2024)*

## EDUCATION



**2022 Bachelor**

Hanoi Univ. of Science and  
Technology, Vietnam



**2024 Master**

Nagoya University, Japan



**2027 PhD (Expected)**

Nagoya University

### **Nagoya University, Nagoya, Japan**

PhD Candidate, Graduate School of Informatics, Department of Intelligent Systems

### **RIKEN, Kyoto, Japan**

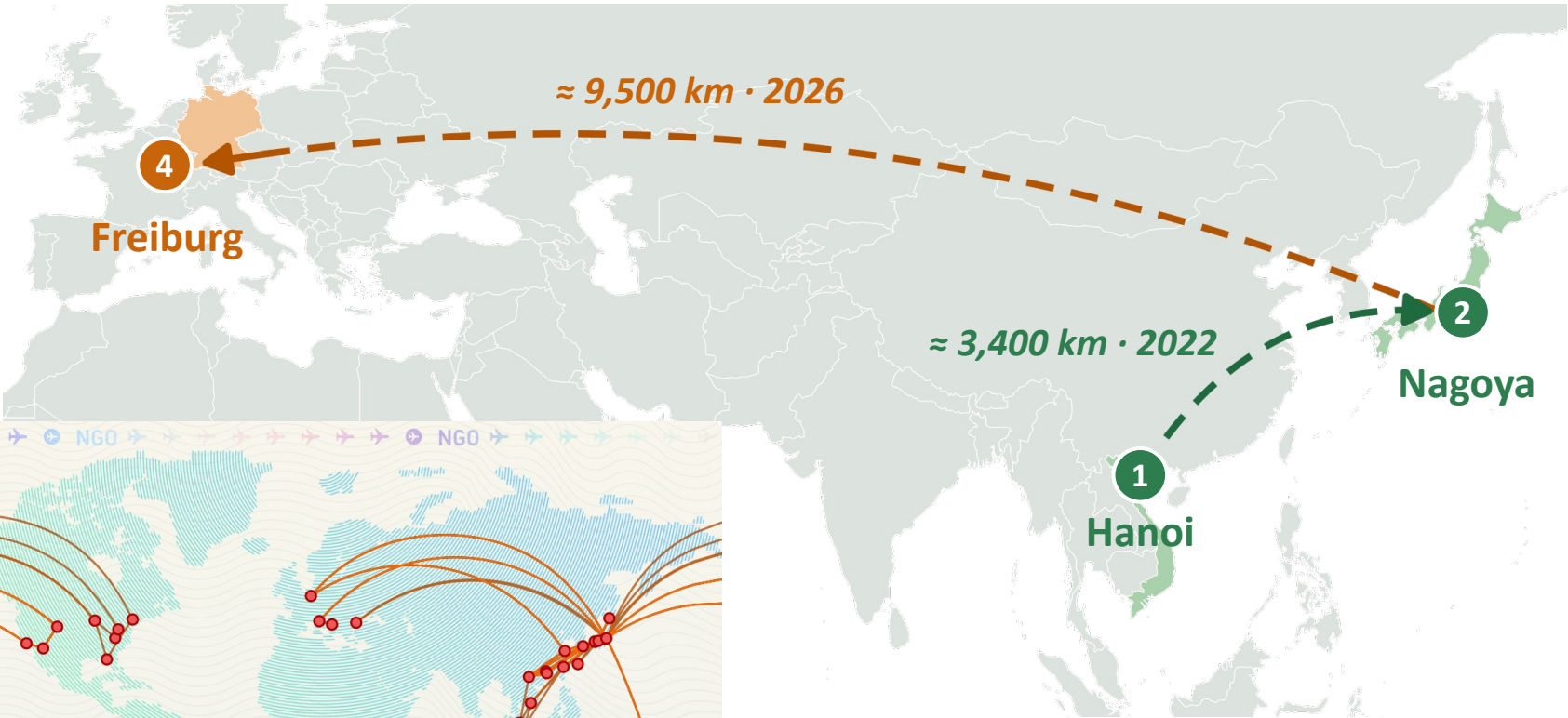
- Student Trainee
- Guardian Robot Project
- Multimodal Data Recognition Research Team

### **University of Freiburg, Germany**

- Rising Star Fellow
- Cluster of Excellence Future Forests
- Host PI: **Prof. Teja Kattenborn**

ABOUT ME

# From Hanoi, to Nagoya, to Freiburg

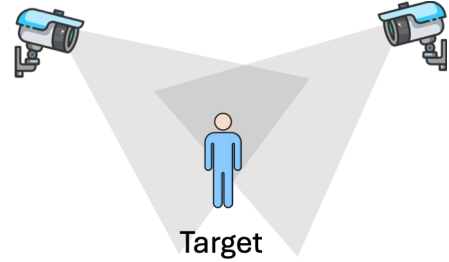


**Nagoya University  $\leftrightarrow$  RIKEN GRP**  
 $\approx 120 \text{ km}$



## ABOUT ME

# My research at a glance



PhD · NAGOYA & RIKEN

### Human Actions from Multiple Sensors

**#Keyword:** Action Recognition, Multi-view, Multi-modal, Open-vocabulary



UNIVERSITY of FREIBURG

### 3D Forest Point Clouds 3dtrees.earth

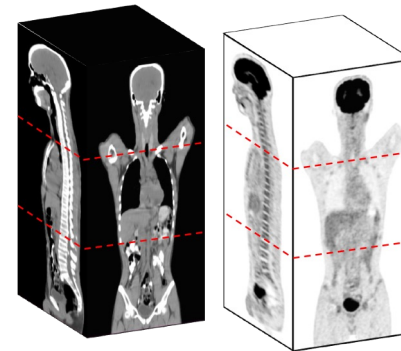
**#Keyword:** 3D Trees, 3D Point Clouds, Instance/Semantic Segmentation



UCLA, USA

### 4D Point Cloud Representations

**#Keyword:** 4D Point Cloud, Representation Learning



AI4LIFE · HUST, VIETNAM

### Medical Imaging and VLMs

**#Keyword:** Diffusion Model, Report Generation, CT/PET Imaging

## BACKGROUND

# Forests are 3D, and ecology needs them tree by tree

# 31%

of Earth's land surface is covered by forest



### Biomass and carbon

Stem and crown volume of each tree



### Crown architecture

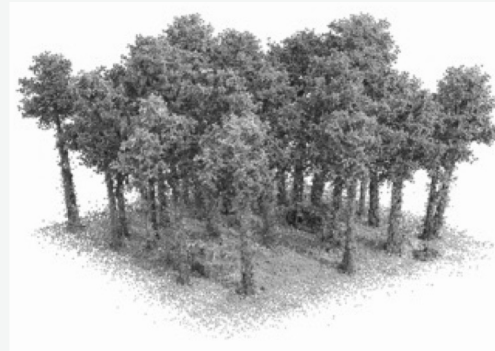
Canopy layers between neighbours



### Health and biodiversity

Species, foliage vs. wood, forest condition

## Joint Semantic + Instance Segmentation



Terrestrial,  
Drone LiDAR Point Cloud

One label Per-tree,  
+ Ground / Wood / Leaf

*Every downstream trait starts from an accurate  
per-tree mask*

## BACKGROUND

# Forest LiDAR is much harder than indoor 3D scenes



### Massive Scale

Hundreds of millions of points per hectare at high acquisition density



### Irregular Density

Occlusion and sensor geometry leave the upper canopy and far stems sparse



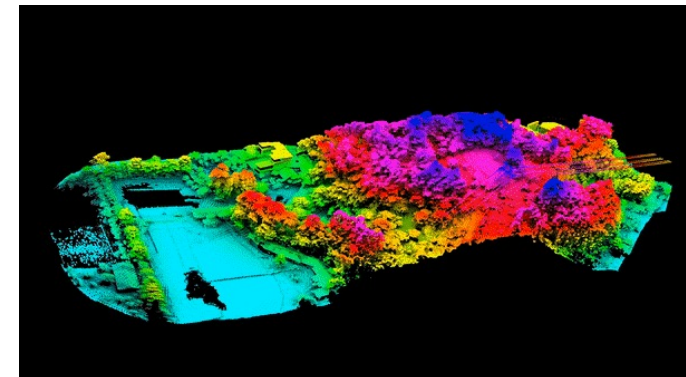
### Entangled Canopies

Overlapping crowns make neighbouring trees hard to separate



### Class Imbalance

Leaf points dominate;  
Ground and Wood are much rarer



## BACKGROUND

# Existing methods and what is still missing

### Clustering-based

*ForAINet · TreeLearn · SegmentAnyTree*

- Sparse-conv U-Net + mean-shift / DBSCAN grouping
- Hand-crafted grouping breaks in dense canopies

### Query-based

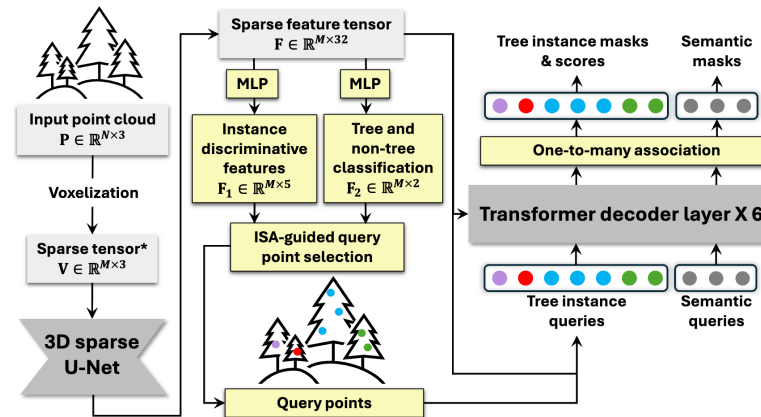
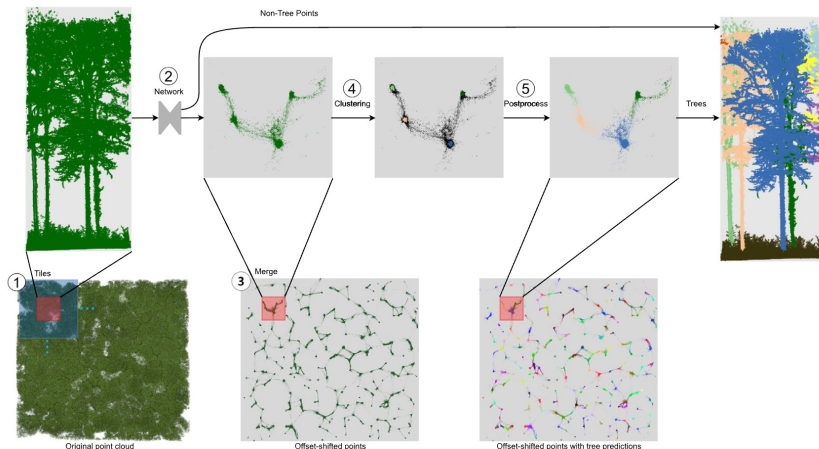
*OneFormer3D · ForestFormer3D*

- Transformer-based: Global attention + mask queries
- State-of-the-art accuracy

### Three Gaps

- 1 Inference Cost**  
Quadratic attention: slow and memory-hungry on large scenes
- 2 No Forest Priors**  
Generic queries ignore canopy geometry, miss weak trees
- 3 No User Control**  
All-or-nothing output: cannot pick or fix one tree

- Gaps 1–2 → ForestMamba
- Gap 3 → SelectAnyTree



\* Represents non-empty voxels as a sparse feature tensor, where  $M$  is the number of voxels containing actual points.

## OUTLINE

# Two steps: faster automatic, then interactive

### PART 1

## ForestMamba

BMVC 2026

*Can forest structure make automatic segmentation both more accurate and cheaper?*

- 1 Vertical-priority sparse Mamba encoder
- 2 Treetop-seeded queries
- 3 Linear-time Mamba query decoder



### PART 2

## SelectAnyTree

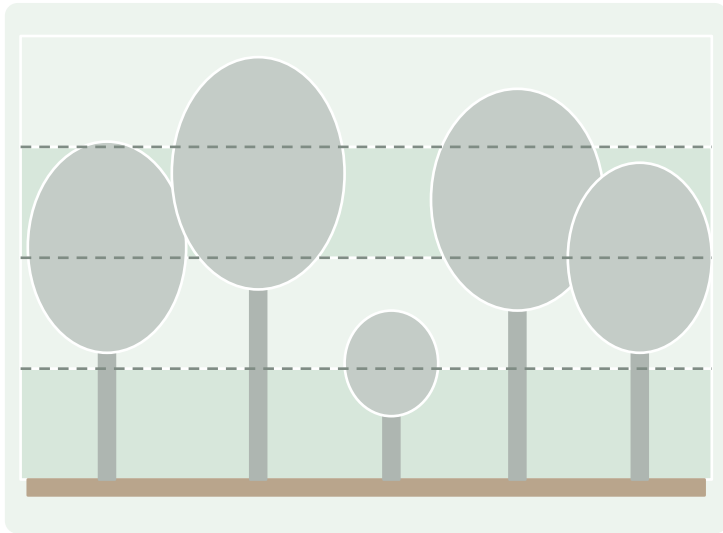
Under Review

*Can a user select, and correct, any single tree with a few clicks?*

- 1 Encode the scene once, reuse for every click
- 2 Click-to-query prompt encoder
- 3 Treetop-seeded as a free first click

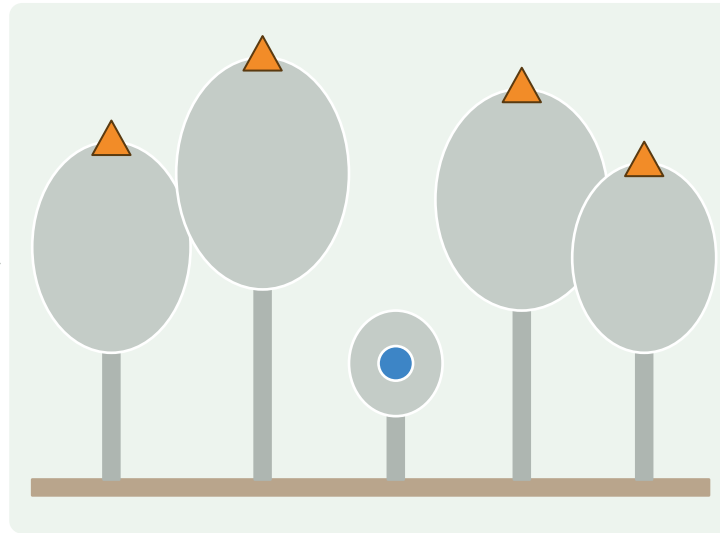
# ForestMamba: the idea

**In:** LiDAR points (x, y, z)    **Out:** every tree, plus ground / wood / leaf



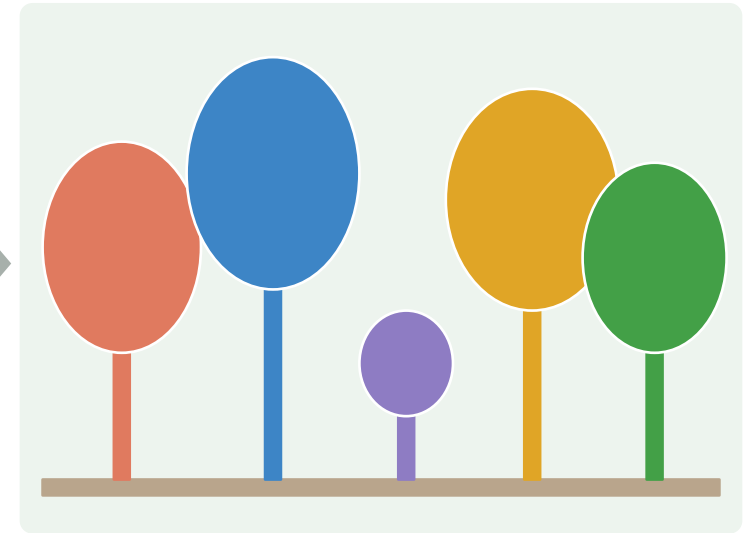
## 1 Read the Forest

Layer by layer, from the ground up



## 2 Mark the Treetops

One marker per tree, plus extra ones for small trees



## 3 Grow Each Tree

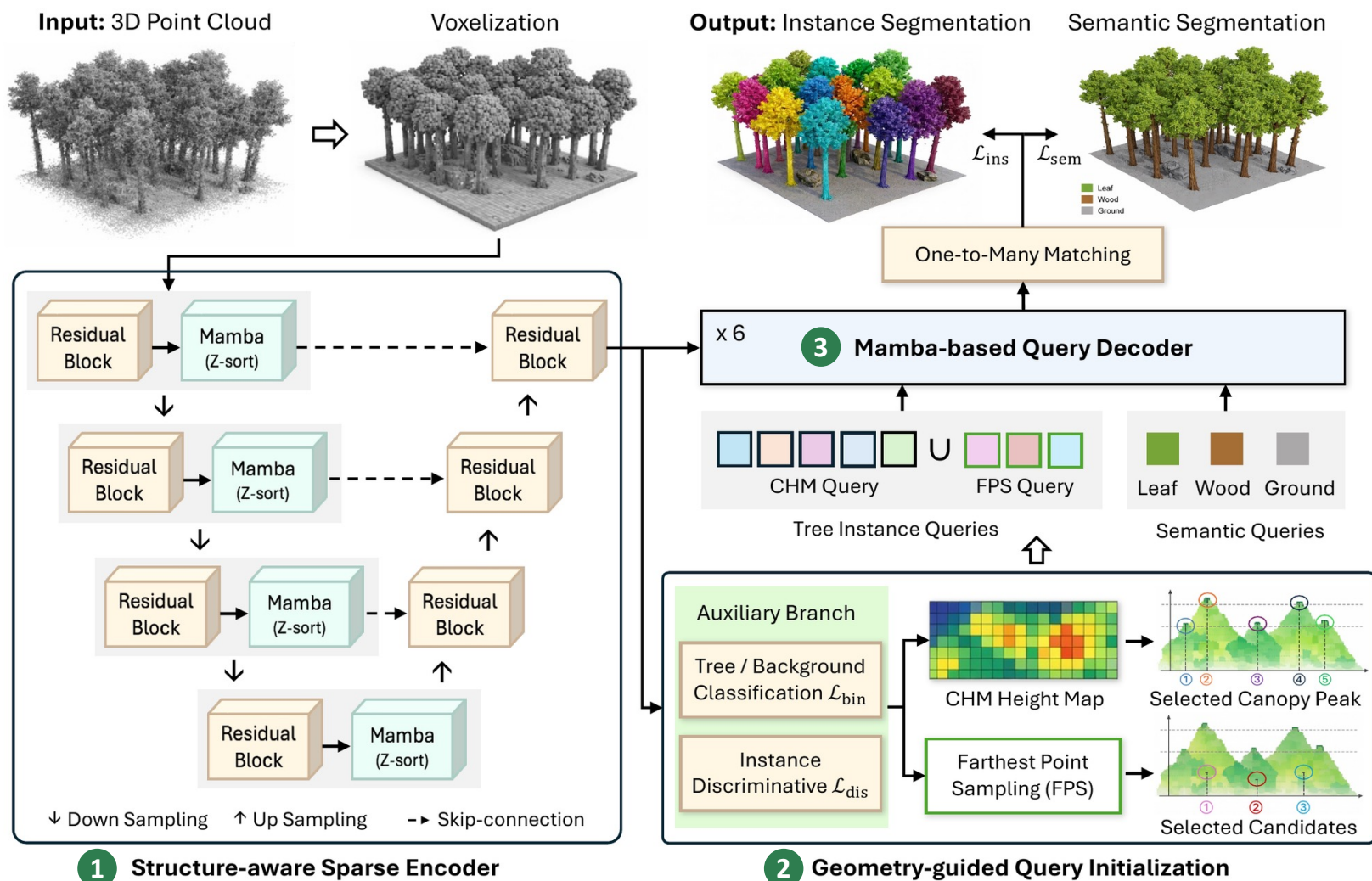
Each marker collects the points of its tree



### In Plain Words

ForestMamba uses two things every forester knows: Trees grow upward, and every tree has a top

# ForestMamba



## 1 Sparse Encoder

Sparse residual blocks + Mamba;  
voxels ordered ground-to-canopy

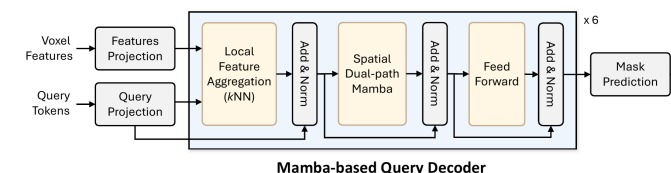
## 2 Geometry-guided Query Initialization

Auxiliary branch (tree / background)  
→ CHM peaks + FPS queries

## 3 Mamba-based Query Decoder

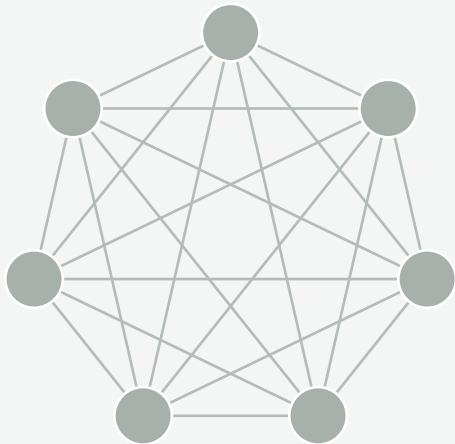
× 6 layers; one-to-many matching;  
outputs trees + ground / wood / leaf

### INSIDE 3: ONE DECODER LAYER



# Why Mamba?

## Transformer: Everyone talks to everyone



Like a meeting where every person talks with every other person

2× bigger plot  
→ 4× more work

Used by the previous best models: accurate, but slow and memory-hungry on large plots

## Mamba: Pass a note down the line

*The note is passed along →*



Like reading a book once while taking notes

2× bigger plot  
→ 2× more work

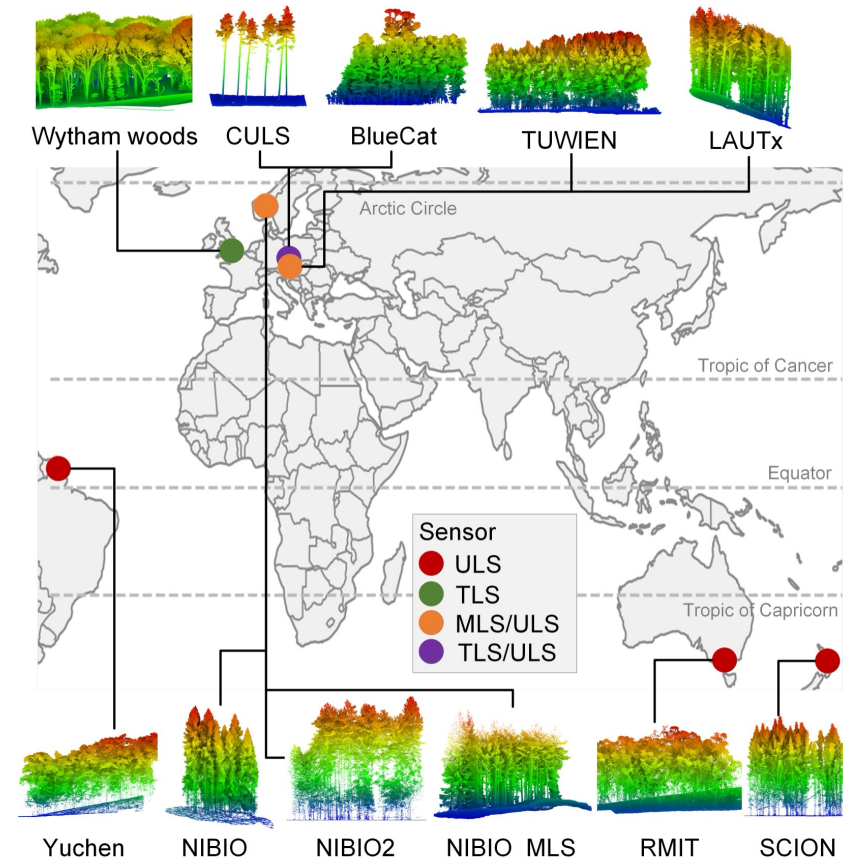
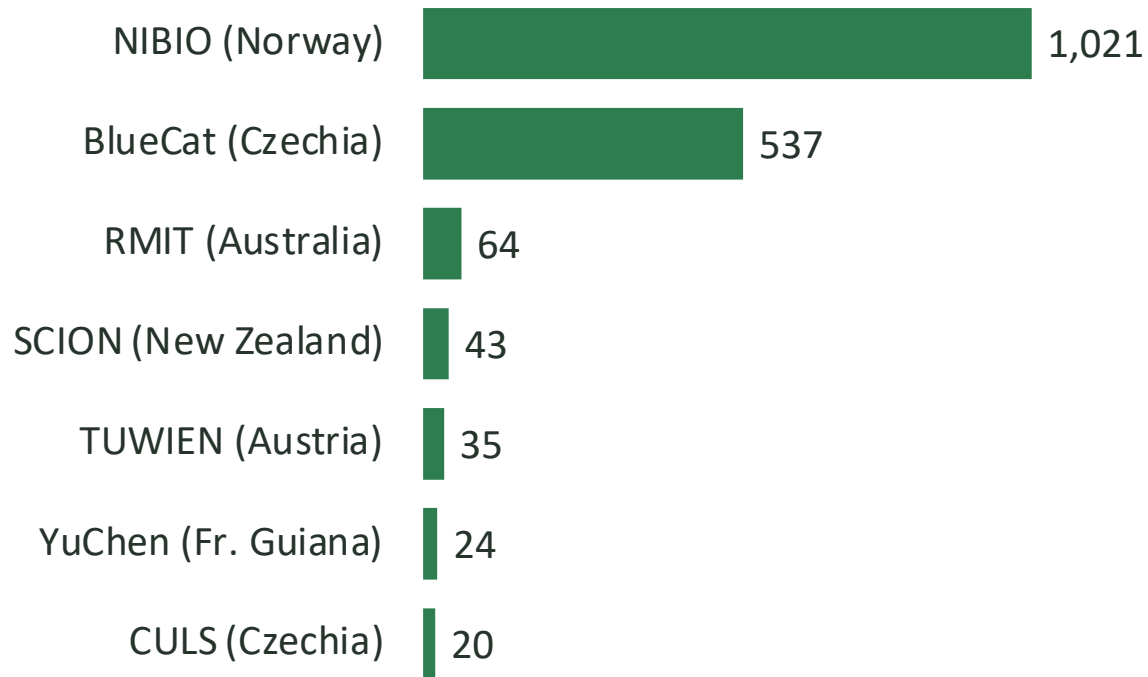
One condition: the reading order must make sense. For trees, we read from the ground up



**In plain words:** Mamba keeps the model fast even for very large forest plots

# How we test it: seven forests, four continents

Test trees per plot region (FOR-instanceV2)



**In plain words:** Seven very different forests – four continents, from boreal to tropical  
Evaluation metrics: Precision, Recall, F1, Coverage, and mIoU

# Quantitative results

| Method             | P           | R           | F1          | Cov.        | mIoU        |
|--------------------|-------------|-------------|-------------|-------------|-------------|
| TreeLearn          | 82.0        | 36.6        | 50.6        | 52.2        | —           |
| ForAINet           | 86.7        | 65.8        | 72.8        | 63.9        | 82.0        |
| OneFormer3D        | 82.2        | 67.3        | 74.0        | 63.8        | 80.8        |
| ForestFormer3D     | 91.3        | 74.9        | 82.3        | 70.5        | 82.1        |
| <b>ForestMamba</b> | <b>91.4</b> | <b>76.6</b> | <b>83.4</b> | <b>71.9</b> | <b>82.3</b> |

*P = Precision; R = Recall; Cov. = Coverage*

**+1.7**

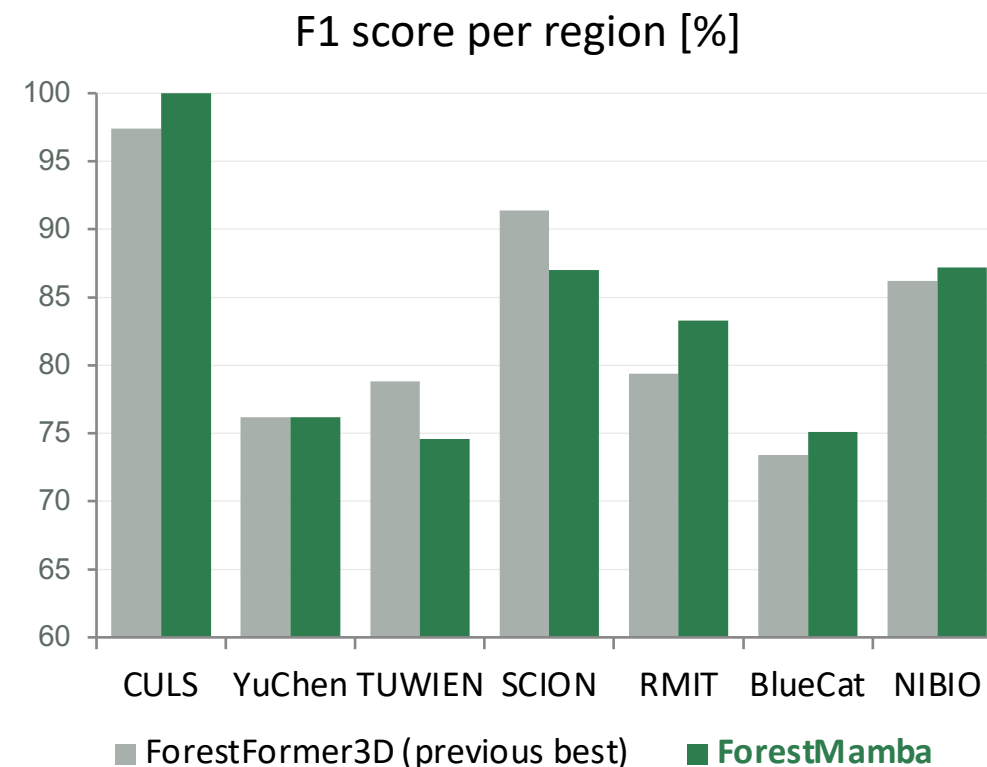
Recall (R): more  
real trees found

**+0.1**

Precision (P): no  
extra false trees

**+1.1**

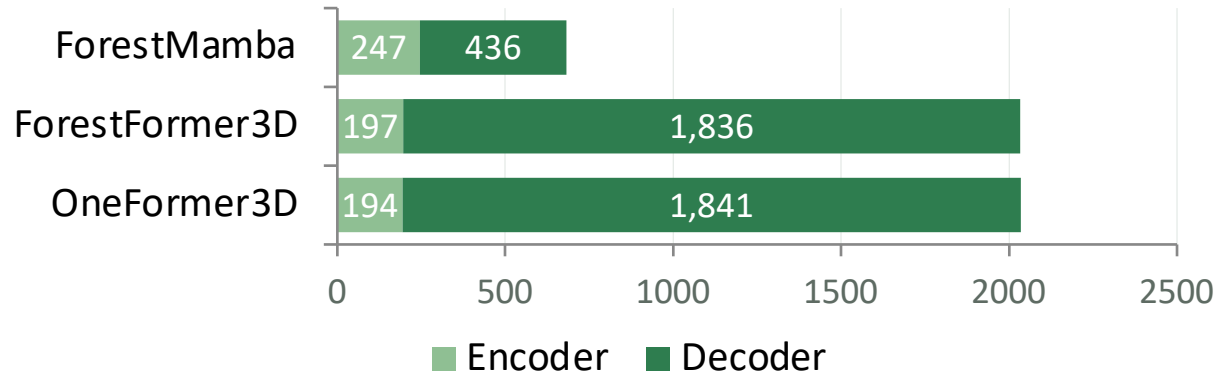
F1 vs. best previous  
model



**In plain words:** ForestMamba finds more of the real trees without adding false ones  
Gains are largest in big, dense plots; two small plots (TUWIEN, SCION) are slightly worse

# About 3× faster, with less than half the memory

Time per plot [ms] on one RTX A6000 GPU



GPU memory [GB]



**2.0 s → 0.7 s**

time per plot

**28 → 12 GB**

GPU memory

**4.2×**

faster tree outlining

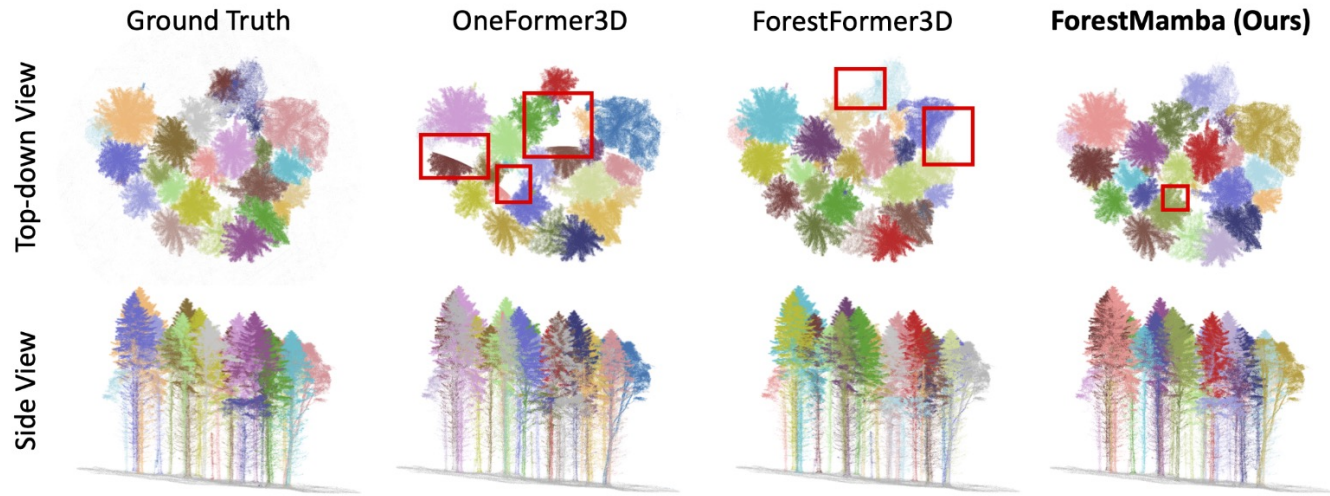
**same size**

17.3M vs 17.5M  
parameters

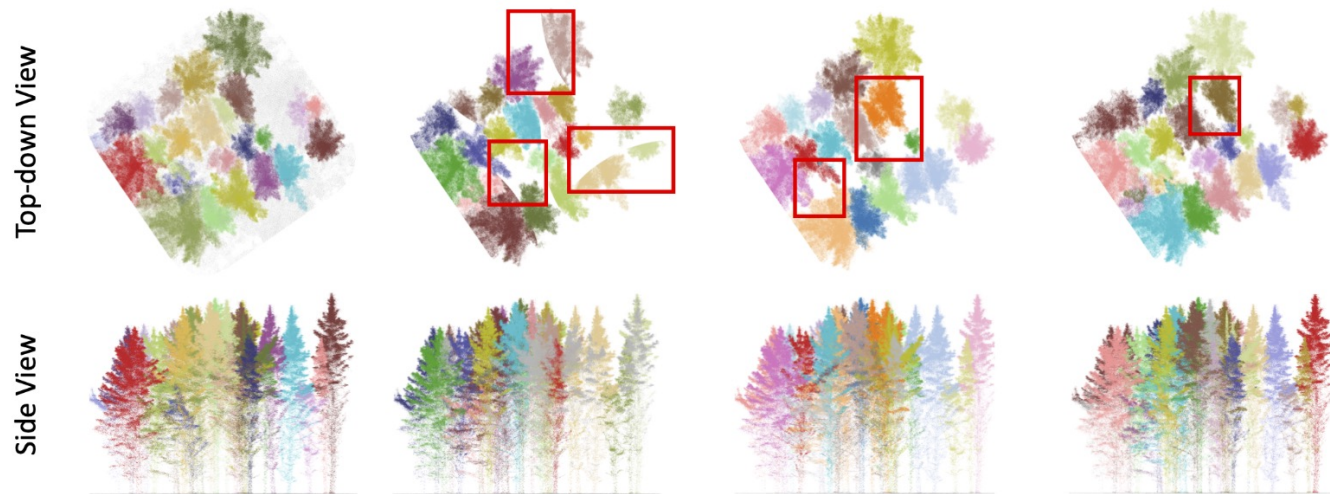


**In plain words:** On the same GPU, a plot that took about 2 seconds now takes 0.7 seconds, and it needs 12 GB instead of 28 GB of GPU memory

# Qualitative results



(a) NIBIO (Norway).



(b) SCION (New Zealand).

## What to look for

- **Red boxes:** missing or fragmented trees
- **Previous models:** several errors per scene in dense crowns and partly hidden trees
- **ForestMamba:** one small error per scene, recovers missing trees in top-down view
- **Side view:** stems and crowns are segmented as one coherent instance

# Automatic is not enough for annotation



## A fixed set of masks

Even at 83 F1, a user cannot pick one tree or repair a wrong one; errors mean manual editing or retraining



## Labels are the bottleneck

A hectare has millions of points and hundreds of crowns; per-tree annotation is slow expert work

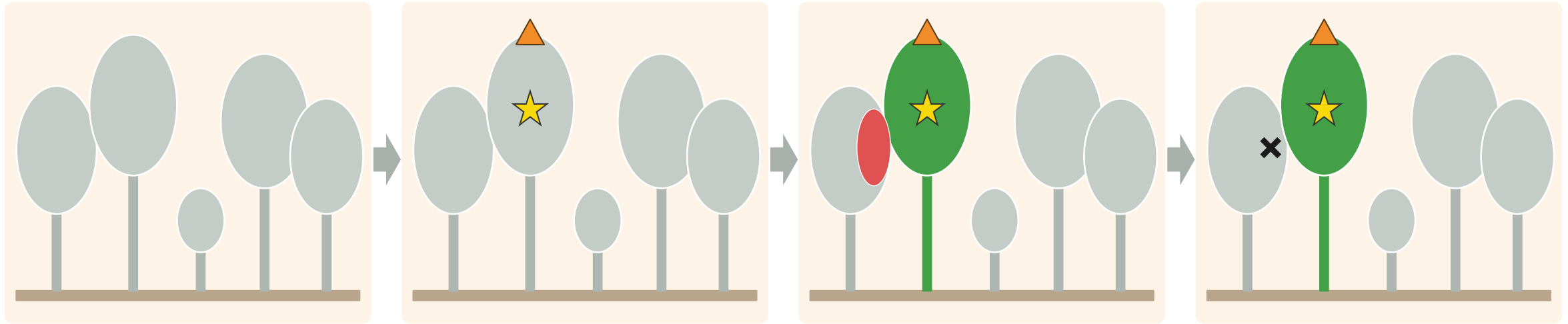


## Promptable segmentation

3D versions exist, but for indoor objects, shapes, and parts, not forests

*Can a user select, and correct, any tree with a few clicks?*

# SelectAnyTree: the idea



- 1 Load the plot**  
read once,  
kept in memory
- 2 Click a tree**  
its treetop (▲); optional
- 3 See the tree**  
here: a bit too much (red)
- 4 Fix with a click**  
× = “not this part”

★ click = “this tree”

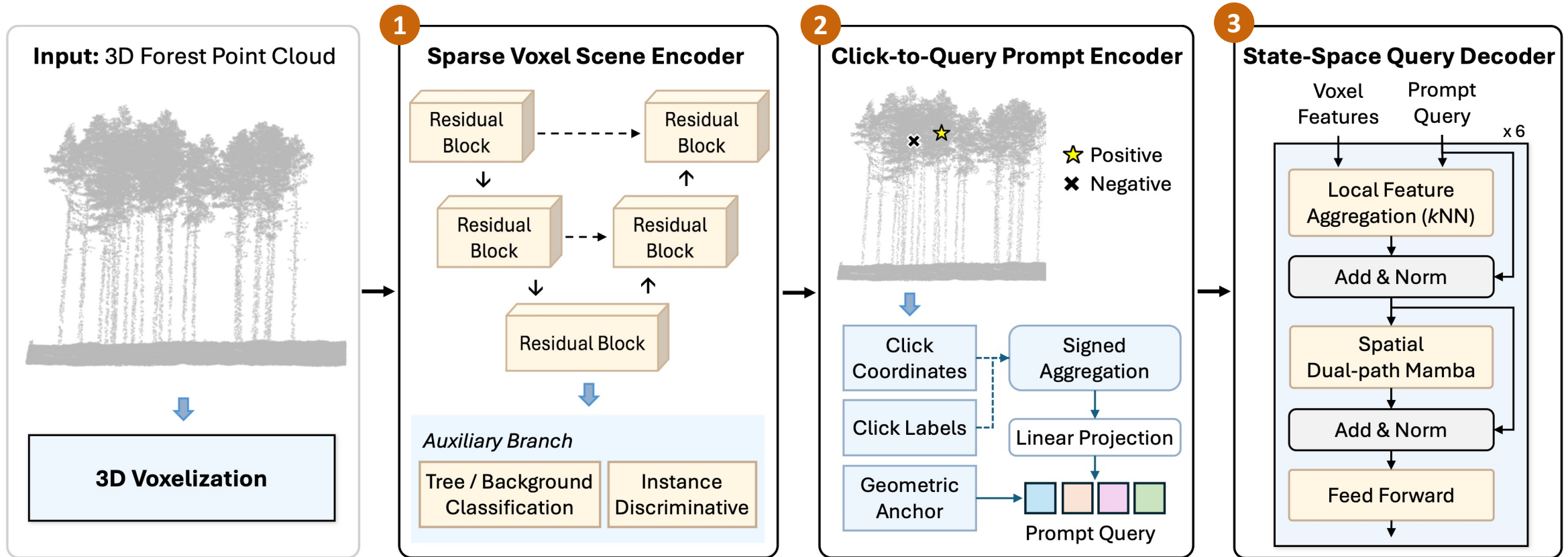
× click = “not this part”

each click gives an answer in a fraction of a second



**In plain words:** It works like the “magic wand” in photo editing, but for single trees in a 3D forest

# SelectAnyTree: from clicks to one tree mask



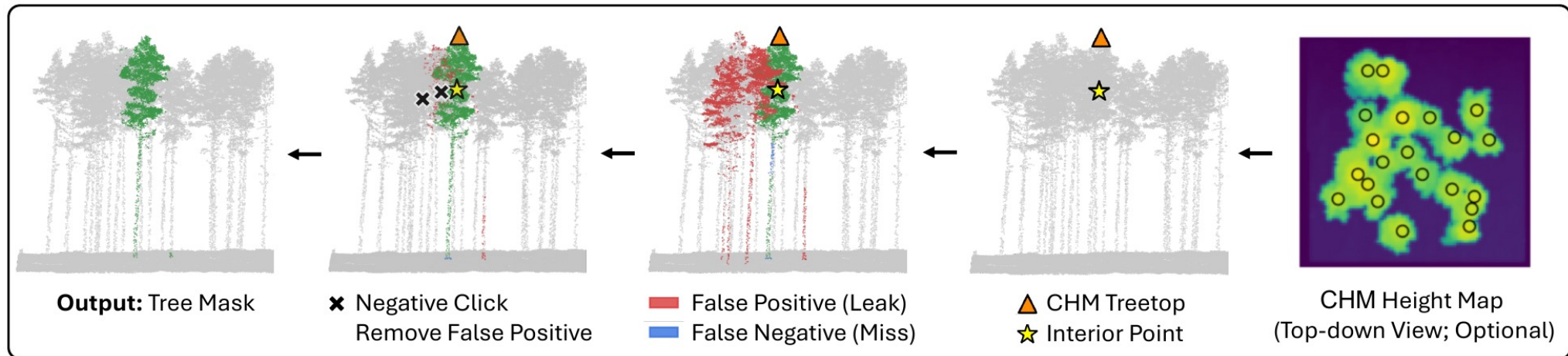
**Task** Point cloud  $P$ , clicks  $C \rightarrow$  Tree mask

① Load the plot once  $\rightarrow$  ② The click become prompt (**Main of SelectAnyTree**)  $\rightarrow$  ③ See the tree

# How we test it: a program plays the annotator

- 1 Click the tree centre**  
Plus the treetop (Optional)
- 2 Look at the answer**  
Compare it with the real tree
- 3 Click the biggest mistake**  
★ for a missed part, × for a wrong part; up to 5 clicks

## A EXAMPLE (READ FROM RIGHT TO LEFT)



**In plain words:** The program always clicks exactly where the model is most wrong  
Real people are less precise, so these are best-case numbers

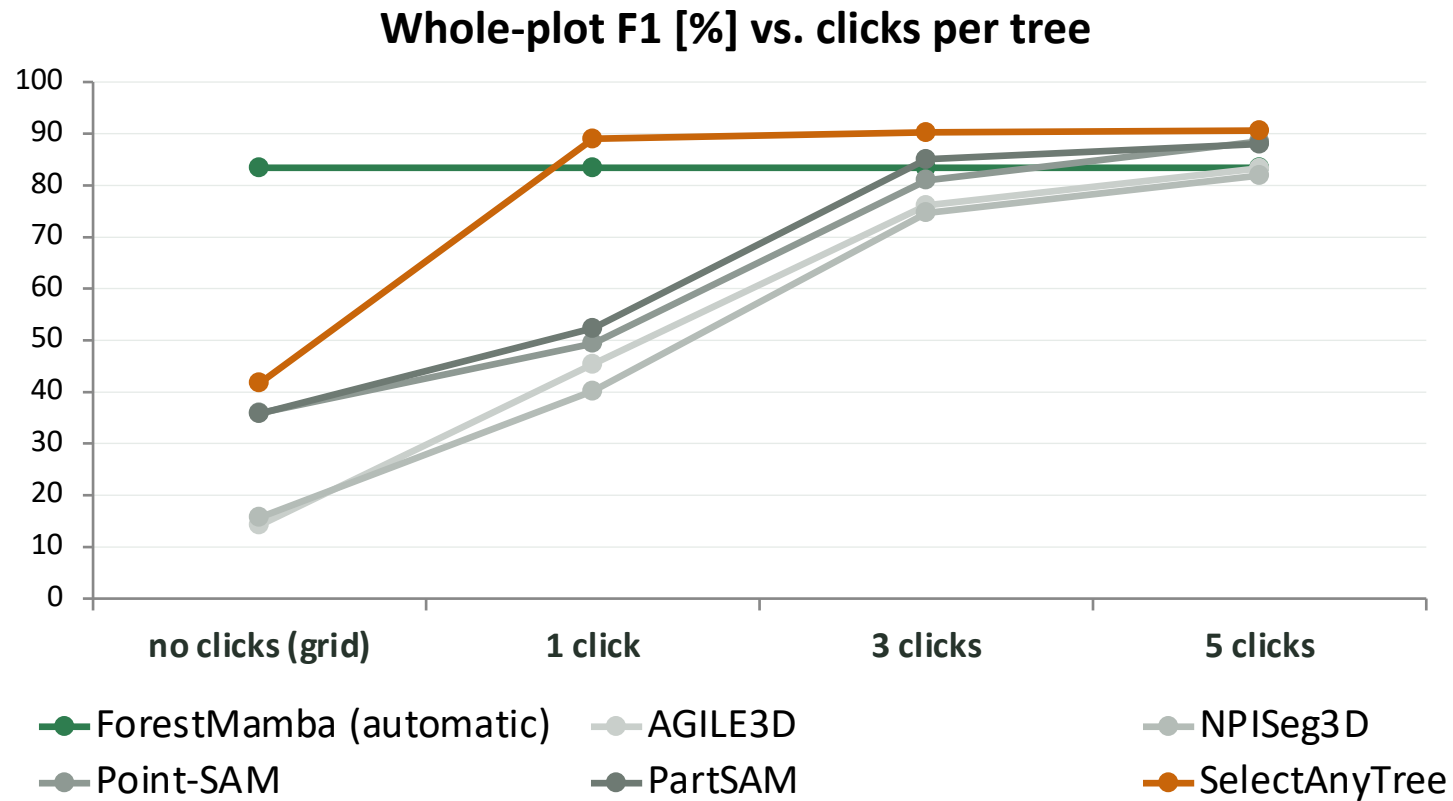
# One click → about 80% overlap with the real tree

| Method                | In-distribution |      |      |             |     |     | Cross-dataset Generalization |      |      |             |     |     | CHM helps? |
|-----------------------|-----------------|------|------|-------------|-----|-----|------------------------------|------|------|-------------|-----|-----|------------|
|                       | IoU @ Clicks ↑  |      |      | NoC @ IoU ↓ |     |     | IoU @ Clicks ↑               |      |      | NoC @ IoU ↓ |     |     |            |
|                       | 1               | 3    | 5    | 0.7         | 0.8 | 0.9 | 1                            | 3    | 5    | 0.7         | 0.8 | 0.9 |            |
| AGILE3D [36]          | 43.8            | 64.6 | 71.9 | 4.6         | 5.9 | 7.8 | 40.8                         | 62.0 | 68.7 | 5.2         | 6.4 | 8.3 | ✗          |
| + w/ CHM              | 43.7            | 59.3 | 66.7 | 5.3         | 6.5 | 8.3 | 40.6                         | 56.0 | 62.5 | 5.7         | 7.0 | 8.8 |            |
| NPISeg3D [14]         | 36.9            | 61.0 | 68.5 | 5.0         | 6.3 | 8.1 | 38.3                         | 59.3 | 65.1 | 5.3         | 6.9 | 8.8 | ✓          |
| + w/ CHM              | 46.4            | 63.9 | 68.3 | 4.8         | 6.2 | 8.2 | 42.6                         | 61.7 | 69.3 | 5.1         | 6.5 | 8.5 |            |
| Point-SAM [38]        | 53.4            | 74.8 | 80.6 | 3.4         | 4.4 | 6.1 | 54.5                         | 75.0 | 81.2 | 3.3         | 4.5 | 6.5 | ✓          |
| + w/ CHM              | 55.2            | 75.1 | 81.6 | 3.4         | 4.4 | 6.0 | 55.7                         | 76.2 | 81.5 | 3.2         | 4.5 | 6.5 |            |
| PartSAM [39]          | 48.8            | 71.2 | 78.3 | 3.7         | 4.8 | 6.6 | 31.4                         | 53.9 | 63.9 | 5.9         | 7.1 | 8.5 | ✓          |
| + w/ CHM              | 53.1            | 74.0 | 80.8 | 3.6         | 4.7 | 6.8 | 44.8                         | 66.1 | 75.5 | 4.3         | 5.8 | 7.8 |            |
| SelectAnyTree (Prop.) | 77.9            | 81.3 | 81.1 | 3.0         | 3.9 | 5.5 | 76.5                         | 80.3 | 81.6 | 2.7         | 3.7 | 6.0 | ✓          |
| + w/ CHM              | 79.9            | 81.9 | 82.4 | 2.9         | 3.8 | 5.4 | 77.7                         | 81.5 | 82.2 | 2.6         | 3.4 | 5.7 |            |



**In plain words:** With one click, SelectAnyTree outlines about 80% of the tree correctly  
Other methods reach about 55%

# One click per tree beats the best automatic model



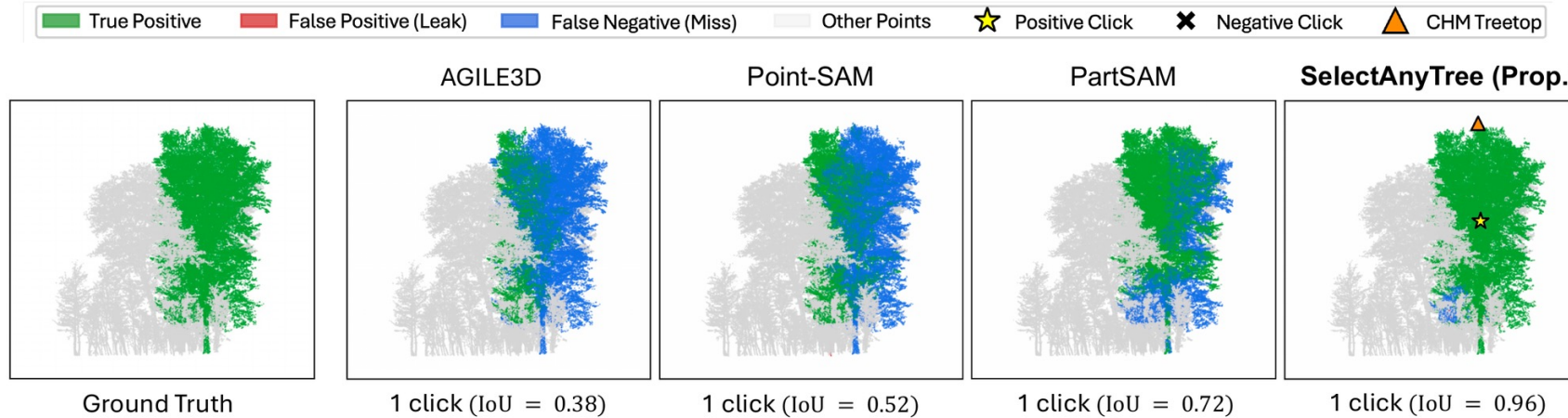
**In plain words:** Without a person, use the automatic model, **ForestMamba**  
If someone can give each tree one click, **SelectAnyTree** gives a better result (89.0 F1)

# Smallest and fastest promptable model

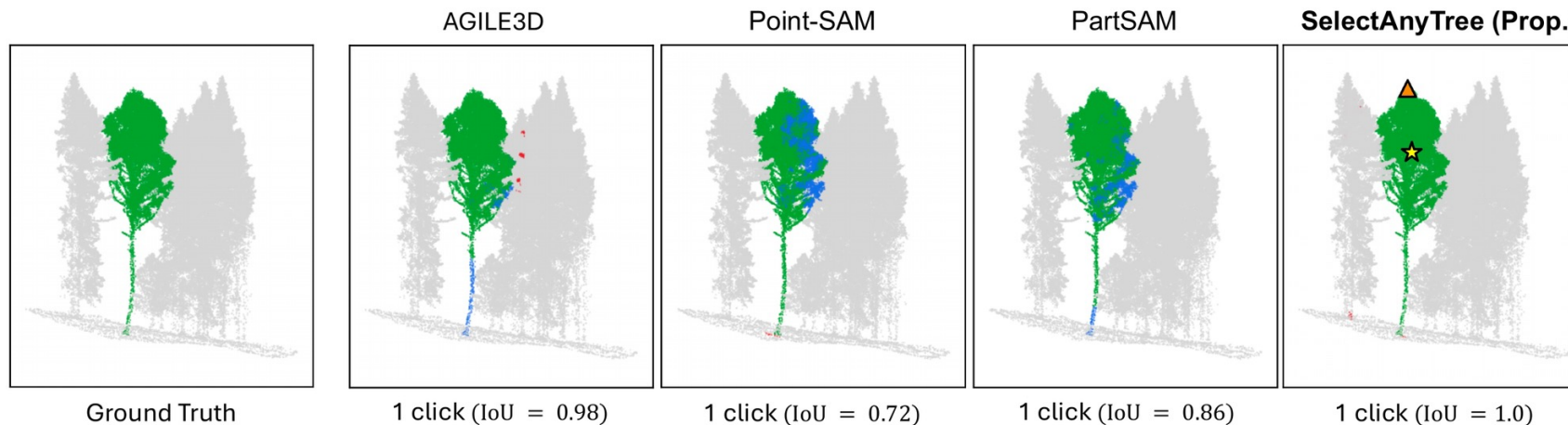
| Method                       | Params<br>[M] ↓ | Time [ms] @ Clicks ↓ |                  |                  | Peak GPU<br>[GB] ↓ |
|------------------------------|-----------------|----------------------|------------------|------------------|--------------------|
|                              |                 | 1                    | 3                | 5                |                    |
| AGILE3D [36]                 | 39.3            | 412 ± 241            | 1,828 ± 653      | 3,286 ± 1,043    | <b>1.0</b>         |
| NPISeg3D [14]                | 40.0            | 249 ± 131            | 1,307 ± 386      | 2,353 ± 664      | 3.1                |
| Point-SAM [38]               | 311.0           | 1,612 ± 602          | 5,062 ± 2,035    | 8,587 ± 3,472    | 22.4               |
| PartSAM [39]                 | 225.0           | 1,549 ± 581          | 2,558 ± 986      | 3,562 ± 1,423    | 2.4                |
| <b>SelectAnyTree (Prop.)</b> | <b>19.4</b>     | <b>122 ± 98</b>      | <b>298 ± 196</b> | <b>571 ± 184</b> | <b>1.9</b>         |
| + w/ CHM                     | <b>19.4</b>     | <b>164 ± 159</b>     | <b>280 ± 156</b> | <b>549 ± 126</b> | <b>1.9</b>         |

**19.4M****Parameters**16× fewer than  
Point-SAM (311M)**571 ms****For 5 clicks**4× faster than NPISeg3D,  
15× faster than Point-SAM**1.9 GB****Peak GPU memory**12× less than Point-SAM;  
AGILE3D uses 1.0 GB

# Qualitative results



(a) BlueCat (Czech Republic) [34]: A dense temperate forest with strongly overlapping crowns.



- **BlueCat (dense):** baselines miss large parts (blue); SelectAnyTree reaches 0.96 IoU
- **NIBIO (conifers):** all methods are closer; only SelectAnyTree gets 1.0
- **Two cues:** the treetop (▲) anchors the apex, the click (★) grounds the stem

SUMMARY

# Two methods that work together

|                       | ForestMamba                                   | SelectAnyTree                                                   |
|-----------------------|-----------------------------------------------|-----------------------------------------------------------------|
| In one sentence       | Finds every tree and its parts automatically  | Let pick and fix any tree with a click                          |
| Forest knowledge used | Trees grow upward; treetops mark trees        | Treetops mark trees (Optional)                                  |
| Key result            | 83.4 F1 · about 3× faster<br>2.3× less memory | About 80% overlap from 1 click<br>89.0 F1 with 1 click per tree |
| Good for              | Large automatic forest inventories            | Making training labels, fixing errors                           |
| Status                | BMVC 2026 · on 3Dtrees.earth                  | Under Review (3DV)                                              |



**In plain words:** ForestMamba for a fast, automatic first pass  
SelectAnyTree to check and correct individual trees in a few clicks